

Assignment 4 Lab

2023-05-31

```
library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.2.3
## Warning: package 'ggplot2' was built under R version 4.2.3
## Warning: package 'tibble' was built under R version 4.2.3
## Warning: package 'tidyr' was built under R version 4.2.3
## Warning: package 'readr' was built under R version 4.2.3
## Warning: package 'dplyr' was built under R version 4.2.3
## Warning: package 'forcats' was built under R version 4.2.3
## Warning: package 'lubridate' was built under R version 4.2.3

## — Attaching core tidyverse packages — tidyverse
2.0.0 —
## ✓ dplyr      1.1.2      ✓ readr      2.1.4
## ✓ forcats   1.0.0      ✓ stringr    1.5.0
## ✓ ggplot2   3.4.2      ✓ tibble     3.2.1
## ✓ lubridate 1.9.2      ✓ tidyr      1.3.0
## ✓ purrr     1.0.1
## — Conflicts —
tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the [8];http://conflicted.r-lib.org/conflicted package[8]; to force
all conflicts to become errors

library(openintro)

## Warning: package 'openintro' was built under R version 4.2.3

## Loading required package: airports
## Loading required package: cherryblossom
## Loading required package: usdata

head(fastfood)

## # A tibble: 6 × 17
##   restaurant item      calories cal_fat total_fat sat_fat trans_fat
##   <chr>         <chr>      <dbl>  <dbl>    <dbl>  <dbl>    <dbl>
## 1 restaurant item      calories cal_fat total_fat sat_fat trans_fat
## 2 <dbl>         <dbl>      <dbl>  <dbl>    <dbl>  <dbl>    <dbl>
```

```

## 1 Mcdonalds Artisan G... 380 60 7 2 0
95
## 2 Mcdonalds Single Ba... 840 410 45 17 1.5
130
## 3 Mcdonalds Double Ba... 1130 600 67 27 3
220
## 4 Mcdonalds Grilled B... 750 280 31 10 0.5
155
## 5 Mcdonalds Crispy Ba... 920 410 45 12 0.5
120
## 6 Mcdonalds Big Mac 540 250 28 10 1
80
## # i 9 more variables: sodium <dbl>, total_carb <dbl>, fiber <dbl>, sugar
<dbl>,
## # protein <dbl>, vit_a <dbl>, vit_c <dbl>, calcium <dbl>, salad <chr>
fastfood

## # A tibble: 515 x 17
## restaurant item calories cal_fat total_fat sat_fat trans_fat
cholesterol
## <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl>
## 1 Mcdonalds Artisan ... 380 60 7 2 0
95
## 2 Mcdonalds Single B... 840 410 45 17 1.5
130
## 3 Mcdonalds Double B... 1130 600 67 27 3
220
## 4 Mcdonalds Grilled ... 750 280 31 10 0.5
155
## 5 Mcdonalds Crispy B... 920 410 45 12 0.5
120
## 6 Mcdonalds Big Mac 540 250 28 10 1
80
## 7 Mcdonalds Cheesebu... 300 100 12 5 0.5
40
## 8 Mcdonalds Classic ... 510 210 24 4 0
65
## 9 Mcdonalds Double C... 430 190 21 11 1
85
## 10 Mcdonalds Double Q... 770 400 45 21 2.5
175
## # i 505 more rows
## # i 9 more variables: sodium <dbl>, total_carb <dbl>, fiber <dbl>, sugar
<dbl>,
## # protein <dbl>, vit_a <dbl>, vit_c <dbl>, calcium <dbl>, salad <chr>
mcdonalds <- fastfood %>%
  filter(restaurant == "Mcdonalds")

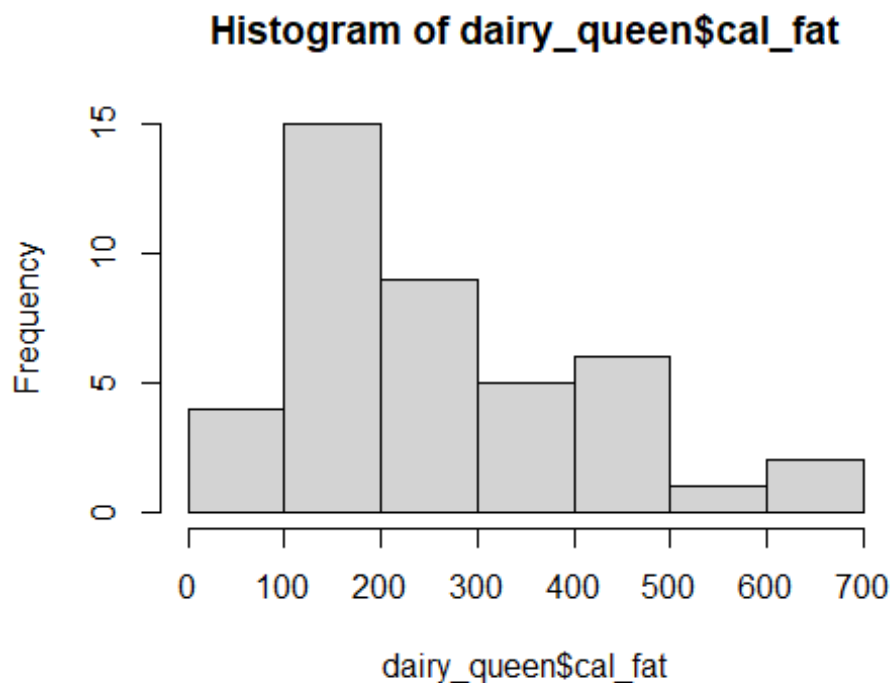
```

```
dairy_queen <- fastfood %>%  
  filter(restaurant == "Dairy Queen")
```

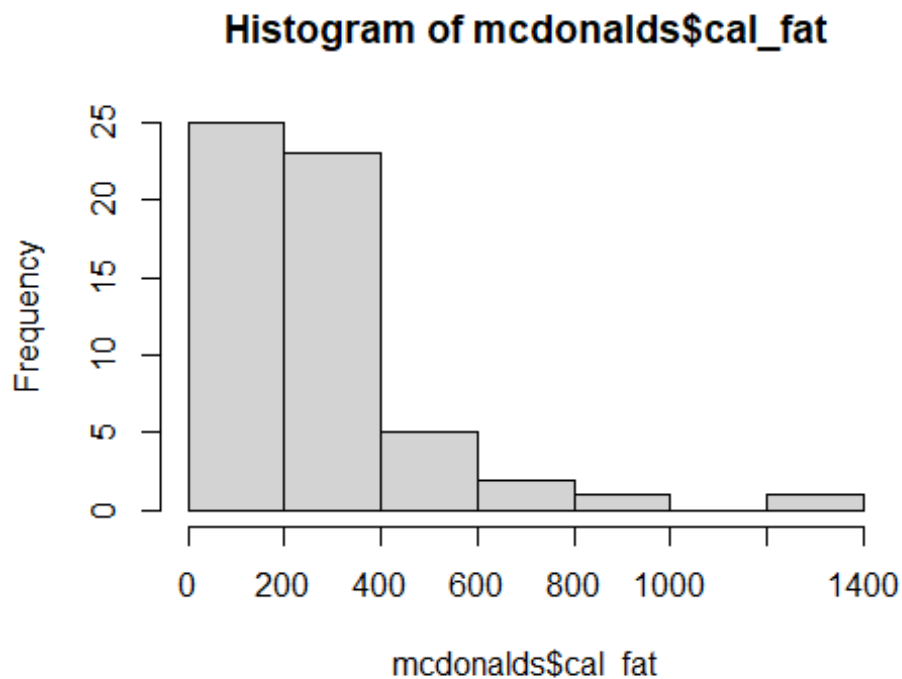
#- Exercise 1: Make a plot (or plots) to visualize the distributions of the amount of calories from fat of the options from these two restaurants. How do their centers, shapes, and spreads compare?

```
dqmean <- mean(dairy_queen$cal_fat)  
dqsd <- sd(dairy_queen$cal_fat)
```

```
hist(dairy_queen$cal_fat)
```



```
hist(mcdonalds$cal_fat)
```



Histogram of McDonald and Dairy queen are both right skewed with differences between them. The mean, mode and range of the mcdonalds fat calorie is greater than dairy queens. Mcdonalds peak (0-200 cal_fat at frequency of 25) Dairy queen peak 9 (100-200 at frequency of 15) The increments increase for mcdonalds is by 200 where as dairy queen is 100.

#- Exercise 2 Based on the this plot, does it appear that the data follow a nearly normal distribution?

```
dqmean <- mean(dairy_queen$cal_fat)
dqsd <- sd(dairy_queen$cal_fat)
```

```
ggplot(data = dairy_queen, aes(x = cal_fat)) +
  geom_blank() +
  geom_histogram(aes(y = ..density..)) +
  stat_function(fun = dnorm, args = c(mean = dqmean, sd = dqsd), col =
"tomato")
```

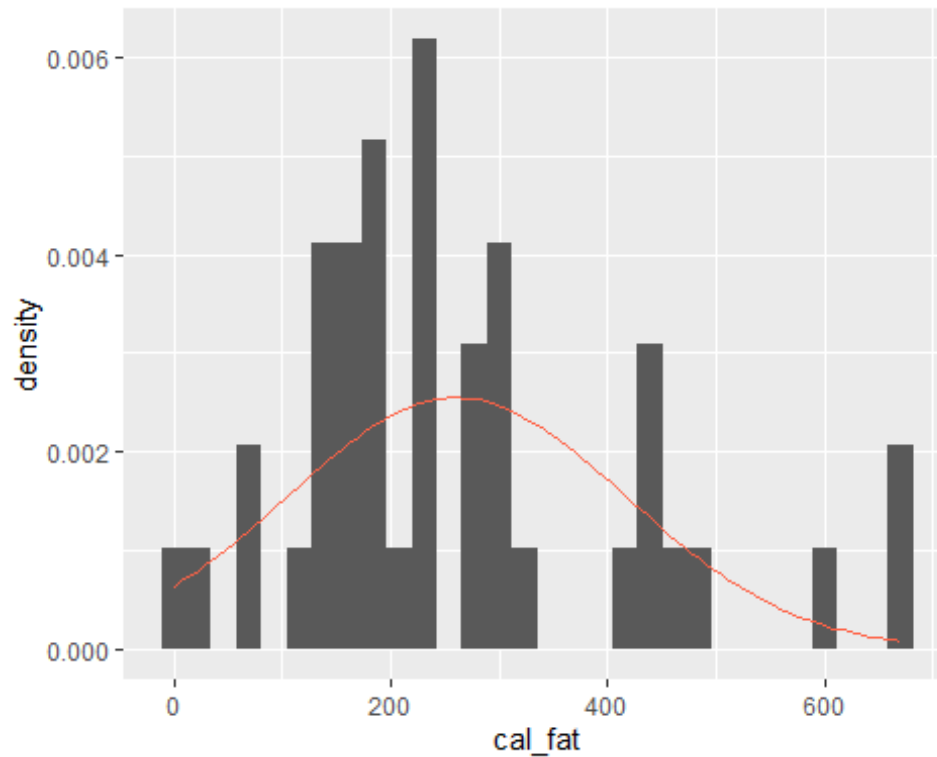
Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.

i Please use `after\_stat(density)` instead.

This warning is displayed once every 8 hours.

Call `lifecycle::last\_lifecycle\_warnings()` to see where this warning was generated.

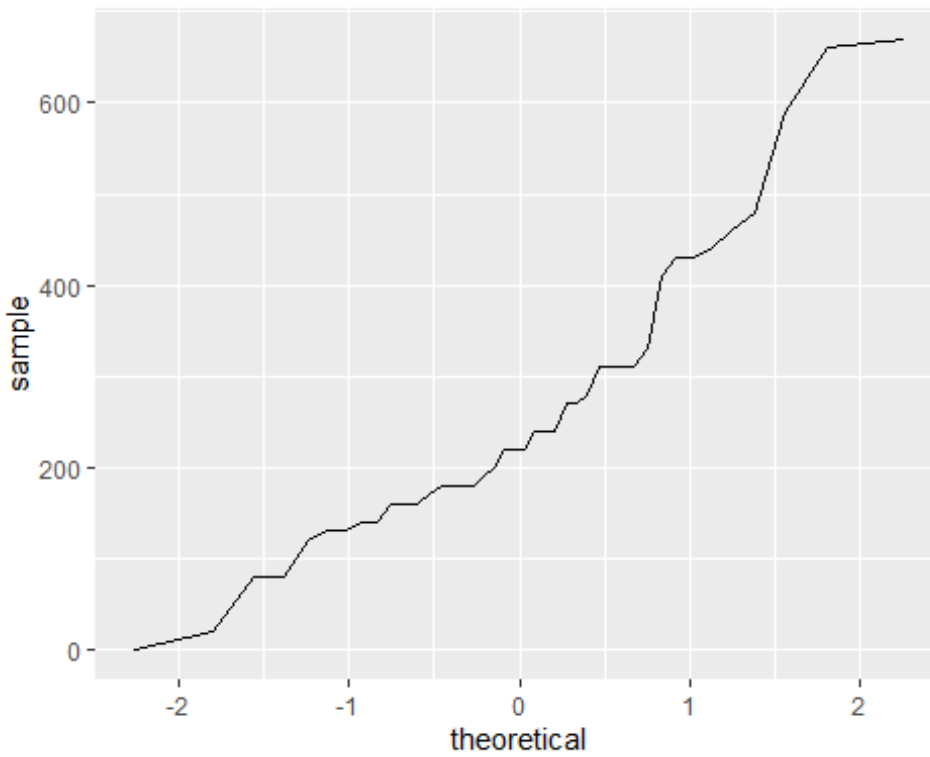
`stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.



The plot somewhat follows the normal distribution plot but the frequencies are much higher than the line (peak crosses the red line). The mean of θ (the centre) is more towards the right and has more density near there (low frequencies at around $+0.5 - 1$ sd). The central peak is around 220 cal_fat

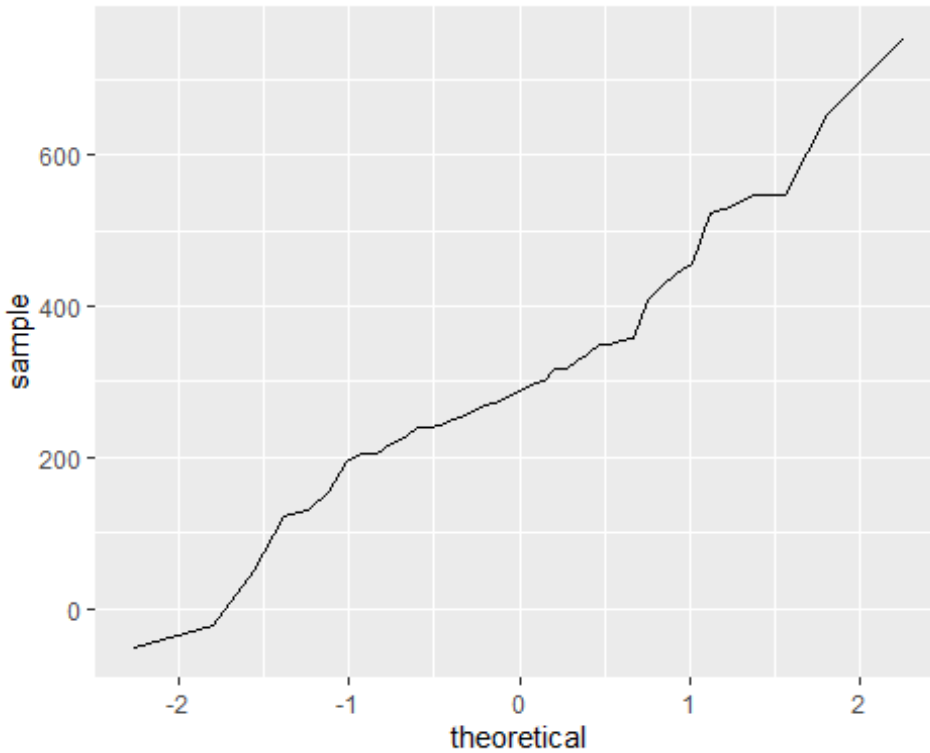
#-Exercise 3: Make a normal probability plot of sim_norm. Do all of the points fall on the line? How does this plot compare to the probability plot for the real data? (Since sim_norm is not a dataframe, it can be put directly into the sample argument and the data argument can be dropped.)

```
ggplot(data = dairy_queen, aes(sample = cal_fat)) +  
  geom_line(stat = "qq")
```



```
sim_norm <- rnorm(n = nrow(dairy_queen), mean = dqmean, sd = dqsd)

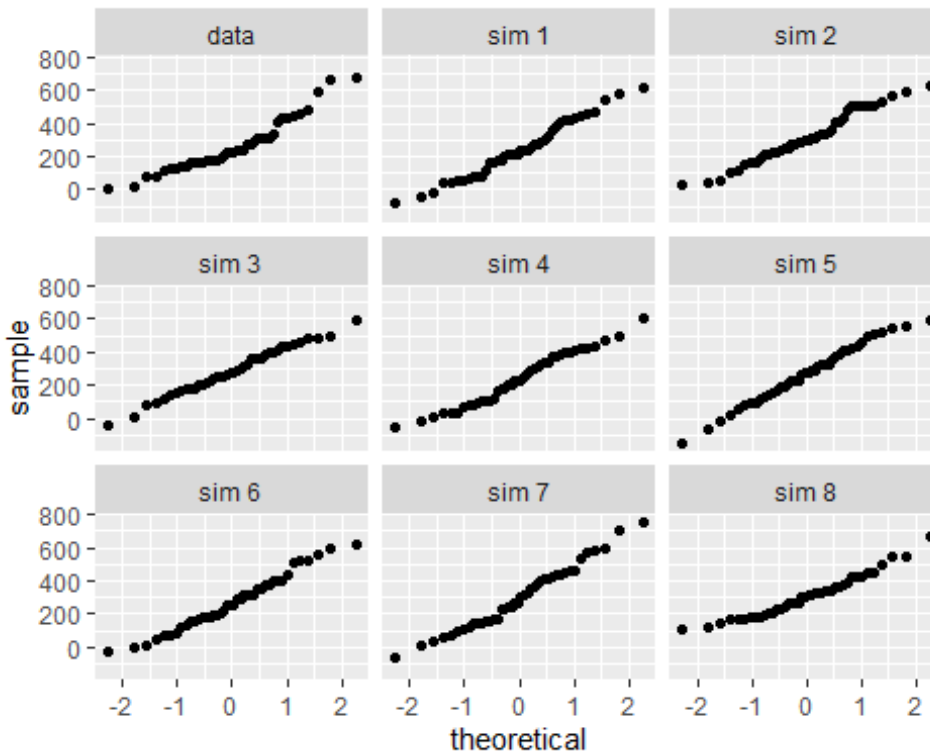
ggplot(data = NULL, aes(sample = sim_norm)) +
  geom_line(stat = "qq")
```



No all the points do not fall on the line and are not exactly the same. The slope of the data points at -2 to -1 are lower and the slope from 1 to 2 is greater.

#- Exercise 4: Does the normal probability plot for the calories from fat look similar to the plots created for the simulated data? That is, do the plots provide evidence that the calories from fat are nearly normal?

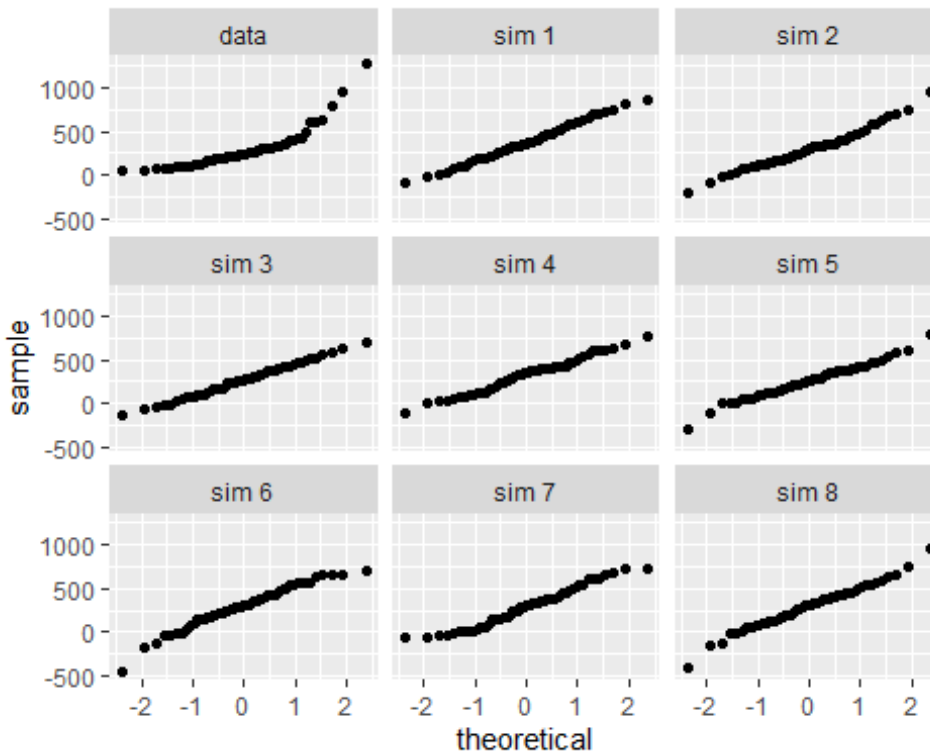
```
qqnormsim(sample = cal_fat, data = dairy_queen)
```



Yes, the plots provide evidence that the calories from fat are nearly normal because the points closely follow a diagonal line and this considered nearly normal and there isn't much deviations.

Exercise 5: Using the same technique, determine whether or not the calories from McDonald's menu appear to come from a normal distribution.

```
qqnormsim(sample = cal_fat, data = mcdonalds)
```



#for most parts the McDonald plot exhibit a nearly normal distribution. Most of the data points for the simulated follow a the diagonal line stating normality and less variance between the data points.

#-Exercise 6: Write out two probability questions that you would like to answer about any of the restaurants in this dataset. Calculate those probabilities using both the theoretical normal distribution as well as the empirical distribution (four probabilities in all). Which one had a closer agreement between the two methods?

```
1 - pnorm(q = 600, mean = dqmean, sd = dqsd)
## [1] 0.01501523

dairy_queen %>%
  filter(cal_fat > 600) %>%
  summarise(percent = n() / nrow(dairy_queen))

## # A tibble: 1 × 1
##   percent
##   <dbl>
## 1 0.0476
```

#Question 1: You are a personal trainer for arnold Schwarzenegger. He eats 7 meals a day. He wants more than 30g of protein per meal to sustain his muscle retention. What is the Likelihood that a randomly selected product from Taco Bell's menu contains over 30 grams of protein?

```
TB <- fastfood %>%
  filter(restaurant == "Taco Bell")

tb_mean <- mean(TB$protein)
tb_sd <- sd(TB$protein)

1 - pnorm(q = 30, mean = tb_mean, sd = tb_sd)

## [1] 0.03891314
```

```
TB %>%
  filter(protein > 30) %>%
  summarise(percent = n() / nrow(TB))

## # A tibble: 1 × 1
##   percent
##   <dbl>
## 1 0.0696
```

#6.9% of the time a product will have over 30g of protein. That is not good for bodybuilders who need to consume around 200g per day. I would stay away. (only good for cheat days)

#Question 2: Mark is trying to lose weight fast! He wants to eat only 250 calories for lunch. What is the probability that a randomly chosen product from Subway is less than 250 calories?

#< 250 cals calculations:

```
subway <- fastfood %>%
  filter(restaurant == "Subway")

sbmean <- mean(subway$calories)
sbsd <- sd(subway$calories)

pnorm(q = 250, mean = sbmean, sd = sbsd)

## [1] 0.1849837

fastfood %>%
  filter(calories < 250) %>%
  summarise(percent = n() / nrow(subway))
```

```
## # A tibble: 1 × 1
##   percent
##   <dbl>
## 1    0.708
```

#70.83% of the time the product will have less than 250 calories. That pretty good if you are on a diet. I would definitely recomend.